Socialized Unlearning: Making Each Other Forget Through Multi-Agent Collaboration

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Advanced Machine Learning Project
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The Problem:

The reasons behind Unlearning are several:

- **LEGAL**: data are affected by privacy issues or copyrights constraint
- **ETHICAL**: data can be biased and create ethical unbalance
- **EPISTEMOLIGIC**: data are useless, obsolete or unwanted for the model

Challenges of unlearning include:

- **Catastrophic forgetting:** Loss of retained knowledge during unlearning.
- **Computational inefficiency:** Retraining the model from scratch most of the time could be impractical.

Why Socialized Learning (SL)?

SL, through **collective collaboration** and **reciprocal altruism**, offers:

- Modular management of knowledge across agents.
- Isolation of learned knowledge for selective forgetting.
- Balancing **stability** and **plasticity**, enabling models to forget without losing versatility.

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Models, tools etc.

We want to apply the “social learning” approach we saw in the paper to perform unlearning. So we are gonna present an alternative version of the code that you can find below:

Socialized Learning: Making Each Other Better Through Multi-Agent Collaboration

Metrics and Benchmarks

We will compare our “social unlearned” models with their normal version in terms of:

Forgetting Effectiveness $$FE = 1 - \frac{\text{Accuracy}_{\text{post-unlearning}}}{\text{Accuracy}_{\text{pre-unlearning}}}$$

Retention Score $$RS = \frac{\text{Accuracy}_{\text{non-target, post-unlearning}}}{\text{Accuracy}_{\text{non-target, pre-unlearning}}}$$

We will compare our “Social Unlearned” models with their different technique unlearning approach models with CIFAR-10/CIFAR-100 datasets.

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Thanks for your attention!

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